

Piyush Singh

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EDUCATION

- ❖ VIT Bhopal, B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning CGPA: 8.8
- ❖ Vanasthali Public School CBSE–Class 12th • Scored 89.6% in CBSE Board Examinations
- ❖ Vanasthali Public School CBSE–Class 10th • Scored 96.4% in CBSE Board Examination

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, Java, HTML, JS, ReactJS

AI and ML skills: ML, DL, Neural Network, GenAI, LLM, Advance RAG, Transformer (HuggingFace), NLP, Computer Vision AI ML

Framework and Tools: LangChain, LangGraph, LangSmith Monitoring, AWS, CrewAI, Ollama, MLflow, DVC, Docker, Vercel, CI/CD, Scikit-learn, OpenCV, NLTK, Git, Docker

Backend Development and APIs: API development using FastAPI and Flask, SQL

Data Science: Python, Statistical Analysis & Modeling, ML, pandas, NumPy Data Visualization using PowerBI, PySpark

PROJECTS

Water Potability Detection: MLOPS and CI/CD https://github.com/PiyushVIT346/water_potability_MLOPS

- Spearheaded end-to-end data pipelines, demonstrating strong **project management** and **problem-solving** skills by reducing data processing time by **30%** and improving model training efficiency.
- Administered the model lifecycle for streamlined efficiency using MLflow and DagsHub, incorporating rigorous tracking, versioning, and a centralized registry, which improved model accuracy by a measurable 8.6%
- Deployed FastAPI-based ML APIs with **CI/CD automation** via GitHub Actions, which **reduced manual deployment**.

Pravdha AI – Legal Assistant : <https://github.com/PiyushVIT346/PravdaAI>

- Engineered Legal AI Assistant using Flask, LangChain, and LangGraph with modular MCP architecture and Blueprint pattern.
- Implemented hierarchical RAG retrieval system: multi-level intent classification, topic categorization, MMR-based dense retrieval, ensemble retrievers, contextual compression with LLM extractors.
- Designed LangGraph state machine for dynamic query routing across specialized FAISS vector stores.
- Built document processing pipeline with sentence-transformers embeddings, multi-query generation, and document compressor pipelines
- Developed secure REST API with SQLite authentication and PDF processing capabilities

Trained Model for CT-Scan, X-Ray and Brain tumor <https://github.com/PiyushVIT346/health-hackathon>

- Built a medical image classifier using CNNs and fine-tuned ResNet-50 to accurately distinguish X-rays, CT scans, and Brain MRIs across heterogeneous datasets, achieving 89% accuracy.
- Developed an NLP-powered disease prediction chatbot that processes user symptom inputs to identify probable conditions and suggest first-aid measures.
- Engineered a rule-based meal plan generator leveraging user health history, dietary preferences, and chronic conditions to deliver personalized nutrition recommendations.

EXPERIENCE

Intern and Trainee (Machine Learning and Data Science) at LearnFlu | Remote | November 2024 – January 2025

- Completed a 3-month internship focused on Python, machine learning, statistics, and data analysis with hands-on training in Pandas, NumPy, Scikit-learn, and Matplotlib.
- Applied practical experience in data preprocessing, feature engineering, model training, evaluation, and deployment across real-world business scenarios.
- Formulated ML solutions including Iranian churn prediction (Logistic Regression, RF, XGBoost), customer segmentation (RFM, KMeans, PCA, DBSCAN), and real estate price prediction using regression techniques.

CDC Summer Internship | Remote | June 2025 – July 2025

- Earned **IBM GenAI Certification:** Built chatbots using NLP and IBM Watsonx.ai and fine-tuned LLMs on IBM Cloud.
- Obtained **Microsoft DP-900 Certification:** Gained foundational knowledge in data, cloud architecture, and deployed ML models using Azure ML Studio; worked with SQL/NoSQL, Synapse, and Power BI for analytics.

AI-Cloud Lab Internship | Remote | January 2026 – Present

- Selected for AIML role and started working on projects till now.
- Completed courses for training purpose: Networking course-IBM and AIML course by Microsoft.

RESEARCH PAPERS

- EMG Signal Disease Detection– Published in JETIR. <https://www.jetir.org/papers/JETIR2507378.pdf>

CERTIFICATION

- Microsoft DP-900: Azure Data Fundamentals
- IBM GenAI Course
- LangChain, LangGraph and Ollama-RAG– Udemy
- MCP: Build App with Claude Langchain Ollama
- Intermediate Machine Learning– Kaggle
- Computer Vision– Great Learning
- Data Structures and Algorithms (DSA)– Simplilearn
- Advanced C++– Simplilearn

Achievement and Honors

- Selected out of 80,000 students from India as mentee for graphs camp by Codeforce Master and learned 17+ advance problem-solving technique in graphs. Solved 1 out of 3 super hard problem on the topic Uber: Bus problem and got featured in Hall of fame.
- Qualified till second - last round in NASSCOM GenAI hackathon. Build AI for Insurance claim just using phone call and most fascinating idea was to eliminate the useless question making it more accessible to both customer and vendor.
- Qualified in prototype round in Health Hackathon.
- Participated in many hackathons: Google Agentic AI (H2Skill exchange hackathon), EY techathon 6.0, Flipkart E-Commerce, MarQing Minds, Elevate-24 Neural Network, HP power Quiz, Adobe India hackathon, TVS credit epic 7. Instead, we got no position, but got new experience that enhance our skills and motivate for further participations.

